Vision Transformer: An Excellent Teacher for Guiding Small Networks in Remote Sensing Image Scene Classification

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***Abstract*— Scene classification is an active research topic in the remote sensing community, and complex spatial layouts with various types of objects bring huge challenges to classification. Convolutional neural network (CNN)-based methods attempt to explore the global features by gradually expanding the receptive field, while long-range contextual information is ignored. Vision transformer (ViT) can extract contextual features, but the learning ability of local information is limited, and it has a large computational complexity simultaneously. In this article, an end-to-end method is exploited by employing ViT as an excellent teacher for guiding small networks (ET-GSNet) in the remote sensing image scene classification. In the ET-GSNet, ResNet18 is selected as the student model, which integrates the superiorities of the two models via knowledge distillation (KD), and the computational complexity does not increase. In the KD process, the ViT and ResNet18 are optimized together without independent pretraining, and the learning rate of teacher model gradually decreases until zero, while the weight coefficient of the KD loss module is doubled. Based on the above procedures, dark knowledge from the teacher model can be transferred to the student model more smoothly. Experimental results on the four public remote sensing datasets demonstrate that the proposed ET-GSNet method possesses the superior classification performance compared to some state-of-the-art (SOTA) methods. In addition, we evaluate the ET-GSNet on a fine-grained ship recognition dataset, and the results show that our method has good generalization for different tasks in terms of some metrics.**

***Index Terms*— Convolutional neural network (CNN), high spatial resolution (HSR) images, knowledge distillation (KD), scene classification, vision transformer (ViT).**

I. Introduction

In recent years, Earth observation (EO) technology has been comprehensively developed, and human beings can be cognitive to the state of the Earth's surface through various modes of remote sensing data [1]–[4]. High spatial resolution (HSR) image is an important and widely used remote sensing data, which can clearly present ground targets and spatial patterns [5]–[7]. Based on HSR images, the prosperity of some real-world applications have been promoted, such as object detection [8], land planning [9], intelligent agriculture [10], traffic analysis [11], and disaster assessment [12].

Scene classification is an active research topic in intelligent interpretation of remote sensing images [13]–[15], and it aims to distinguish a predefined semantic category (e.g., "commercial area," "ground track field," and "mobile home park") for each given image according to its visual information [16]–[18]. As shown in Fig. 1, HSR images possess complex spatial layouts with various types of objects, which always results in characteristics of high intraclass diversity and interclass similarity, and it is necessary to extract scene-level semantic features for accurate classification [19]–[21]. Therefore, various methods are designed in the remote sensing community, and they include handcrafted feature-based methods and deep learning-based methods [6], [7], [13].

Early methods are based on handcrafted features, such as local binary pattern (LBP) features, scale-invariant feature transform (SIFT) features, and histogram of oriented gradients (HOGs) features. Furthermore, it is an effective approach to combine handcrafted features and feature encoding methods, such methods include bag-of-visual-words (BoVW), latent Dirichlet allocation (LDA), and probabilistic latent semantic analysis (pLSA), and high-order statistical information can be extracted for remote sensing scenes [22]–[24]. However, it is clear that designing high-performance handcrafted descriptors is time-consuming and relies heavily on domain knowledge. In addition, handcrafted features can only describe scene information in a single aspect or several aspects. The abovementioned drawbacks restrict these methods to further improve classification results [25].

Deep learning, a novel paradigm for machine learning, can automatically learn high-order abstract features from raw data, and it has been widely explored to perform scene classification in the remote sensing community [26]–[30]. Zou *et al.* [31] adopted a deep belief network (DBN) to achieve feature selection, and an iterative feature learning method was proposed to obtain reliable reconstruction weights. Han *et al.* [32] developed a patch-based spatial-spectral hierarchical convolutional sparse autoencoder (HCSAE), which fully represented high-level semantics of remote sensing scenes. Compared with DBN and sparse autoencoder (SAE), con-

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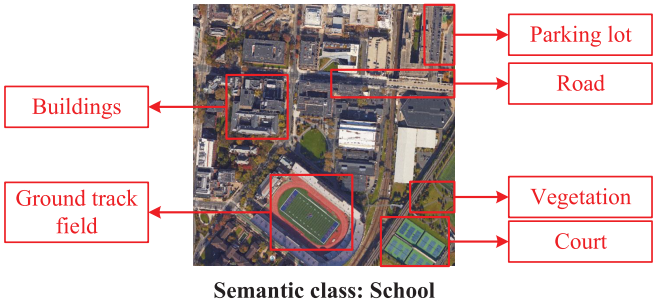


Fig. 1. HSR images possess complex spatial layouts with various types of ground objects.

volutional neural networks (CNNs) are good at capturing local structures and gradually expand receptive fields so that they possess more powerful ability for feature abstraction in image recognition [33]–[35]. In [36], [37], CNNs were newly designed based on the dense connection structure, and they were optimized from scratch. In general, this scheme is flexible for structure construction, and the obtained models perform better than some widely used CNNs (e.g., AlexNet, VGGNet, GoogLeNet, and DenseNet) in the scene classification. However, it is time-consuming and difficult to train high-performance models from scratch because of limited labeled remote sensing data [28], [38], [39].

Considering that the basic elements of the image are the same, it is a good approach to use transferred models that pretrained on large-scale datasets. In [40]–[44], pretrained CNNs were employed as feature extractors, and some encoding methods were introduced to process feature maps from convolutional layers, including locality-constrained linear coding (LLC), improved Fisher kernel (IFK), pLSA, vector of locally aggregated descriptors (VLAD), BoVW, and their variants. In [42], [43], [45], and [46], multilayer features [e.g., convolutional features and fully connected (FC) features] were integrated to improve feature discrimination ability because different layers possess complementary information. In [47] and [48], researchers combined various CNNs to explore the classification performance of fusion models. Furthermore, some novel technologies were also exploited to achieve scene classification, such as graph convolution [46], multiscale pooling [49], covariance pooling [50], long short-term memory (LSTM) [51], and attention mechanism [52]. The above methods have good performance by employing pretrained CNNs as feature extractors, but these transferred models do not consider the characteristics of remote sensing images. Namely, remote sensing images and natural images are quite different in spatial layout, imaging angle, and background information, while the pretraining model ignores these factors [53].

Currently, many methods take pretrained CNNs as a backbone, and some novel modules are designed according to the characteristics of HSR remote sensing images. By using remote sensing data to fine-tune the whole model under small learning rate (LR), more satisfactory results can be obtained in this way. In [54]–[58], many joint loss functions were introduced instead of traditional cross-entropy loss, and authors adopted metric learning regularization (e.g., center loss and its variants) to solve the challenges of high within-class diversity and between-class similarity in the scene classification. In [59]–[62], by imitating the human visual perception, information of crucial local regions was highlighted in HSR scenes, and some advanced technologies were developed, such as multiple instance (MI) learning, local semantic perception module, and weakly supervised key area detection strategy. In [20], [63]–[65], some novel pooling layers were designed to enhance the discriminative ability of CNN activations, such as multilevel improved circle pooling, deep object-centric pooling (DOCP), covariance pooling, and bilinear pooling. Furthermore, some networks, such as CapsNet, generative adversarial network (GAN), and gated recurrent unit (GRU), were combined with CNNs to improve the classification performance [66]–[68]. However, it is difficult for CNNs to capture long-range dependencies, which are important for HSR classification.

Transformer possesses the ability to learn long-range contextual relations, and it has been exploited in the remote sensing community [69]. He *et al.* [70] proposed a spatial-spectral transformer (SST) to mine relationships of sequential spectra in a long distance for hyperspectral image (HSI) classification. He *et al.* [71] developed an HSI-BERT method, in which BERT means bidirectional encoder representations from transformers, and it possessed the ability to capture the global dependence between pixels. In the scene classification, Bazi *et al.* [72] adopted vision transformer (ViT) [73] to explore long-range information, and several data augmentation methods were introduced to boost model discriminative ability. Zhang *et al.* [74] designed a powerful method termed remote sensing transformer (TRS), which has the structure of CNNs and transformers. Deng *et al.* [75] developed a novel framework by integrating CNNs and ViT, and competitive results were achieved in two public datasets. However, the high complexity of time and space limits the application potential of ViT-based methods, and it is necessary to reduce model complexity, while good performance is maintained. Knowledge distillation (KD) [76] is an appropriate strategy to achieve model compression. However, existing methods are based on CNNs as teacher model, which cannot explore contextual information, and the training of teacher/student models is two separate links [77]. In addition, the structure of student model is very simple to pursue low computational complexity, and the performance of the teacher model is far better than that of the student model.

Considering the abovementioned limitations, we employ ViT as an excellent teacher for guiding small networks (ET-GSNet) in the remote sensing image scene classification. The main contributions of this article are summarized as follows.

1) The ET-GSNet method employs ViT as a teacher model to extract rich long-range contextual information, which is transferred to a small student model (i.e., ResNet18) via KD. In this way, the student model combines the advantages of the two models, and model complexity does not increase.
2) In the ET-GSNet method, the student and teacher models are optimized together. In the training stage, the LR of teacher model gradually decreases until zero, while

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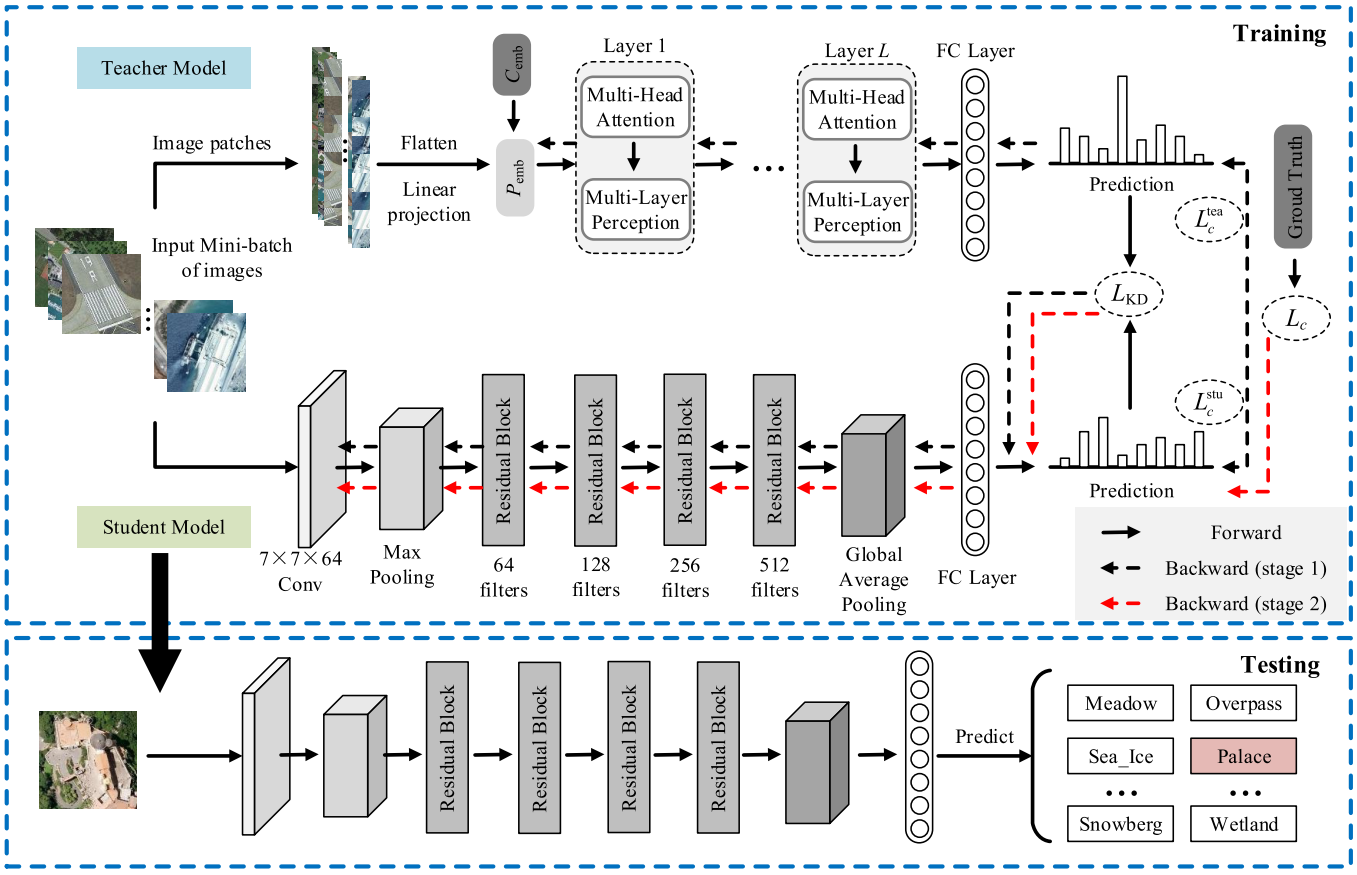


Fig. 2. Framework of the proposed ET-GSNet method. (P_{emb}: position embedding, C_{emb}: class embedding, FC: fully connected, Conv: convolution, L_{KD}: KD loss, L_c: cross-entropy loss, L_c^{stu}: cross-entropy loss of student model, and L_c^{tea}: cross-entropy loss of teacher model.)

the weight coefficient of KD loss is doubled. Based on the above procedures, dark knowledge from the teacher model can be transferred to the student model more smoothly.

3) The proposed ET-GSNet method possesses good generalization in multiple research topics. Experiments on the tasks of scene classification and fine-grained ship recognition show that the student model in the ET-GSNet achieves excellent results in terms of several widely used metrics.

The rest of this article is organized as follows. Section II introduces the details of our proposed ET-GSNet method. Section III discusses the experimental results organized on five public datasets, including four scene classification datasets and a fine-grained ship recognition dataset. Section IV concludes this article, and future work is prospected.

II. Proposed ET-GSNet Method

HSR images are characterized by complex spatial layout and ground object structure, and it is an effective approach to further boost the classification performance via exploring long-range spatial relations. Inspired by the above fact, the ET-GSNet method is developed to fully excavate long-distance contextual information and local structure features of HSR images via KD, and the framework of our proposed method is shown in Fig. 2. As shown in Fig. 2, the ET-GSNet method involves two components, including teacher model and student model. Due to the strong ability of learning long-range spatial relations, ViT is introduced as a teacher model to explore the relationship between HSR image patches in the ET-GSNet. Compared to ViT, ResNet18 [78] with small parameters is adopted as a student model to learn the local structure information. For the phase of training, teacher and student models are optimized together with two stages to transmit long-distance dependency knowledge from ViT to ResNet18. Furthermore, the loss function of two stages and the associated KD loss coefficient are both different. In Sections II-A–II-C, the teacher network, the student network, and KD are clarified in detail.

A. Teacher Network

ViT is capable of extracting long-range dependency relations between sequences through multihead attention, and it is proven to have outstanding performance on image classification. Therefore, ViT is acted as a teacher model in the proposed method, and it passes on knowledge to the student model.

For each input HSR image $I_{in} \in \Re^{H\times W\times 3}$ with spatial dimensions $H \times W$, it is cropped into 2-D patches, and all patches are flattened to be a sequence $S_{eq} \in \Re^{N\times(P_r^2\cdot 3)}$, in which $N = H \times W/P_r^2$ and P_r denote the number of 2-D patches and the resolution of each patch, respectively.

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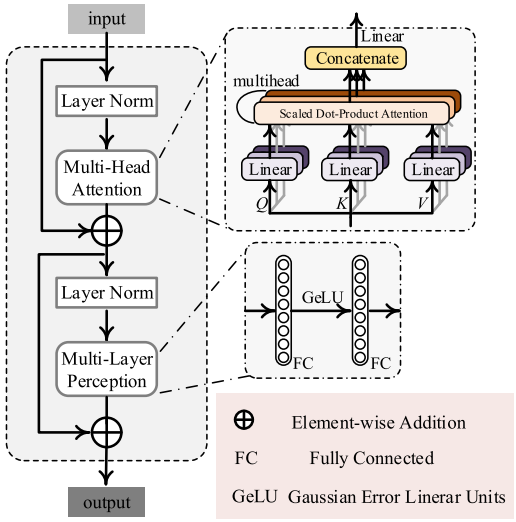


Fig. 3. Detailed structure of the lth layer in the ViT model. (Q, K, and V are feature vectors.)

Then, an embedding matrix $M_e \in \Re^{(P_r^2 \cdot 3) \times D}$ is introduced for linear projection, and patch sequence S_{eq} is mapped into the D-dimensional space. Subsequently, learnable class embedding C_{emb} is concatenated with the embedded patches, and it performs as the final image description of teacher model. Considering the necessity of recording each patch location, position embedding $P_{\text{emb}} \in \Re^{(N+1) \times D}$ is further added to generate the input representation R_0 of the ViT model. Hence, the input of teacher model is formulated as follows:

$$R_0 = \left[C_{\text{emb}}; S_{\text{eq}}^1 M_e; S_{\text{eq}}^2 M_e; \ldots; S_{\text{eq}}^N M_e\right] + P_{\text{emb}}. \tag{1}$$

As the most important module of ViT, transformer encoder is stacked of L layers with the same structure, which is composed of alternating blocks of multihead self-attention (MSA) and multilayer perception (MLP). It is worth noting that the layer norm (LN) is applied before each block, and residual connection is used to convey information of HSR images. As shown in Fig. 2, the formed input R_0 is processed by transformer encoder, and long-range dependency information of HSR images is highlighted via multihead attention. For the lth layer, the intermediate variable R_l' and the output R_l are defined as

$$R_l' = \text{MSA}(\text{LN}(R_{l-1})) + R_{l-1} \quad (l = 1, \ldots, L) \tag{2}$$

$$R_l = \text{MLP}\left(\text{LN}\left(R_l'\right)\right) + R_l' \quad (l = 1, \ldots, L). \tag{3}$$

In the processing of classification, the first element of last layer R_L^0 with LN is considered as the presentation to predict semantic class. Therefore, the final output of teacher model F_{tea} can be obtained by

$$F_{\text{tea}} = \text{LN}\left(R_L^0\right). \tag{4}$$

To further understand the operation of each layer, the detailed structure of the lth layer is shown in Fig. 3, and the crucial part is the MSA block. As shown in Fig. 3, the MSA module has four different layers, including linear layer, self-attention layer, concatenation layer, and linear layer.

Different from traditional attentions, the adopted self-attention maps a set of queries (Q) and a set of key-value (K, V) pairs to output vectors, and the dimensions of the three learnable matrices (i.e., Q, K, and V) are d_k, d_k, and d_v, respectively. For each element in the input sequence R, the values of Q, K, and V are calculated by multiplying R against three learned matrices $\mathcal{U}_{\text{QKV}} \in \Re^{D \times 3d_k}$

$$[Q, K, V] = R\mathcal{U}_{\text{QKV}}. \tag{5}$$

To determine relevance among elements of HSR image patches sequence, dot products are computed the query with all keys coupled with a scaling factor $(1/(d_k)^{1/2})$ and then conducted a softmax function on the values to form attention weights. Therefore, the generated attention weights (W^{att}) implying the key part of image patches are defined as follows:

$$W^{\text{att}} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{6}$$

To reduce sequence computation, it is beneficial to map the queries, keys, and values t times with different linear projections and then perform (6) in parallel. Subsequently, formed t vectors are concatenated via the concatenation layer, and each vector is called a head. After concatenation, the MSA results can be obtained by

$$\text{MSA}(R) = \text{Concat}\left(W_1^{\text{att}}; W_2^{\text{att}}; \ldots; W_t^{\text{att}}\right)W \tag{7}$$

where $W \in \Re^{t \times d_v \times D}$ means learnable weights of a feedforward layer and $\text{Concat}(\cdot)$ denotes the operation of concatenation. It is worth mentioning that $d_k = d_v = D/t$ and $t = 12$ parallel attention heads are employed in this work.

When the output of MSA module is formed, the corresponding results are further processed by the MLP module in Fig. 3. In the MLP module, two FC layers with the Gaussian error linear unit (GELU) are stacked to produce the output of the single layer in the transformer encoder. The equation of GELU is listed as follows:

$$\text{GELU}(x) = x\phi(x) = x \cdot \frac{1}{2}\left[1 + \text{erf}\left(x/\sqrt{2}\right)\right] \tag{8}$$

in which $\phi(x)$ represents the standard Gaussian cumulative distribution and $\text{erf}(x) = (2/\sqrt{\pi}) \int_0^x e^{-\eta^2} d\eta$. Furthermore, the referred LN in the transformer encoder is calculated by

$$\mu^f = \frac{1}{B}\sum_{i=1}^{B} a_i^f \tag{9}$$

$$\sigma^f = \sqrt{\frac{1}{B}\sum_{i=1}^{B}\left(a_i^f - \mu^f\right)^2} \tag{10}$$

where B represents the number of hidden units of a layer, a_i^f denotes the summed inputs of the ith hidden unit in the fth layer, and all hidden units in a layer share the same normalization terms μ and σ. Then, the normalized hidden unit $\hat{a}^f$ can be generated via

$$\hat{a}^f = \frac{a^f - \mu^f}{\sqrt{\left(\sigma^f\right)^2 + \epsilon}} \tag{11}$$

in which ϵ is a constant for numerical stability.

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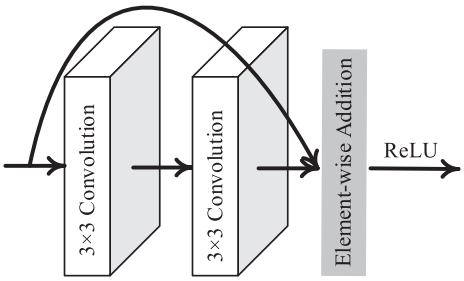


Fig. 4. Detailed structure of the residual block in the ResNet18. (ReLU represents rectified linear unit.)

B. Student Network

As widely used models, CNNs are usually composed of convolutional layer, pooling layer, FC layer, and so on. Benefitted from layer-by-layer structure, abstract features can be gradually extracted from stacked layers. Various CNNs models have been proposed in the past few years, and ResNet is one of the most classic frameworks. Here, ResNet18 is selected as the student model in the ET-GSNet, which is the shallowest structure of ResNet with small parameters. Of course, other small CNNs also have the potential to be integrated into the ET-GSNet method as a student model.

ResNet18 inherits all advantages and characteristics of CNNs, and residual connection is first introduced into ResNet18 to relieve problem of vanishing gradients. Driven by considerably increased depth, ResNet18 further enhances deep representations of HSR images, and more abstract discriminative information of local structure can be refined. As shown in Fig. 2, input HSR images are processed by ResNet18 that consists of a convolutional layer (7×7 kernel), a max-pooling layer, and a series of residual blocks, and then followed by a global average pooling (GAP) layer and an FC layer. The equation of GAP, which converts feature map $U \in \Re^{Y \times Z \times C}$ into feature vector $[g_1, g_2, \ldots, g_C] \in \Re^C$, is defined as

$$g_c = \frac{1}{Y \times Z} \sum_{y=1}^{Y} \sum_{z=1}^{Z} u_{y,z,c} \quad (c = 1, 2, \ldots, C) \tag{12}$$

where Y, Z, and C represent the height, width, and channel of feature maps, respectively. Furthermore, the detailed structure of residual block is given in Fig. 4, and it is composed of two successive convolutional layers with 3×3 kernel accompanied by the rectified linear unit (ReLU). The output of each residual block B_{out} connected by skip connection is calculated by

$$B_{\text{out}} = \text{ReLU}(\text{Convs}(\text{In}_b, \{w\}) + \text{In}_b) \tag{13}$$

in which In_b is the input of the block and $\text{Convs}(\text{In}_b, \{w\})$ denotes the learnable convolutional mapping. Therefore, for each HSR image $I_{\text{in}} \in \Re^{H \times W \times 3}$, the output of student model F_{stu} is formed.

C. Knowledge Distillation

As the mainstream of model compression, KD is dexterously transmitting the knowledge of large neural networks (teacher model) to a small model (student model) and keeping high accuracy. Under the circumstance, the student model trains on hard labels from ground-truth data and also optimizes on the softer labels, which are produced by the teacher model with more potential information.

In the proposed ET-GSNet, KD is acted between ViT and ResNet18 and successfully distilled long-range dependency knowledge from ViT into ResNet18 via optimizing them together. It is worth mentioning that the optimization strategy of ET-GSNet is two-stage training with different α in Fig. 2. Specifically, the student and teacher models simultaneously optimize via minimizing L_{fir} with small coefficient α in the first stage, and then, the LR of teacher model gradually decreases until zero and the student model is solely tuned on the L_{sec} with big coefficient 2α for the second stage. Hence, the loss function for training stage of ET-GSNet has three terms, including KD loss (L_{KD}), cross-entropy loss of teacher model (L_c^{tea}), and cross-entropy loss of student model (L_c^{stu}). The training loss functions L_{fir} and L_{sec} are defined as follows:

$$L_{\text{fir}} = \alpha L_{\text{KD}} + L_c^{\text{tea}} + L_c^{\text{stu}} \tag{14}$$

$$L_{\text{sec}} = 2\alpha L_{\text{KD}} + L_c^{\text{stu}} \tag{15}$$

where α represents the KD loss coefficient that balances the importance of KD loss and cross-entropy losses. In the ET-GSNet, L_{KD} is the key to make ResNet18 imitate probability distribution of ViT, and the definition of L_{KD} is provided as follows:

$$L_{\text{KD}} = T^2 \cdot \text{KLdiv}\left(Q_{\text{stu}}^T, Q_{\text{tea}}^T\right) \tag{16}$$

in which T represents the temperature of KD, which determines the soft degree of labels. $\text{KLdiv}(\cdot)$ means the Kullback–Leibler divergence that can measure the distance between two distributions, and the equation is listed as

$$\text{KLdiv}(p \parallel q) = \sum p(x) \log \frac{p(x)}{q(x)} \tag{17}$$

where $p(x)$ and $q(x)$ are the two probability distributions for discrete random variables.

In (16), Q_{stu}^T and Q_{tea}^T are the class probabilities generated by using a high-temperature softmax function instead of traditional softmax ($T = 1$) for student and teacher models, and the higher value of T produces a softer probability distribution over classes. The jth element of Q_{stu}^T and Q_{tea}^T are denoted as

$$Q_{\text{stu}}^T(j) = \frac{\exp\left(F_{\text{stu}}^j / T\right)}{\sum_b \exp\left(F_{\text{stu}}^b / T\right)} \tag{18}$$

$$Q_{\text{tea}}^T(j) = \frac{\exp\left(F_{\text{tea}}^j / T\right)}{\sum_b \exp\left(F_{\text{tea}}^b / T\right)} \tag{19}$$

where F_{stu}^j and F_{tea}^j are the jth logits produced from outputs of student model F_{stu} and teacher model F_{tea}, respectively. In addition, b means the number of elements in F_{stu} and F_{tea}.

In the developed ET-GSNet, L_{KD} is the bridge of long-range dependency information transmission, and L_c provides hard label information to complement with L_{KD}. Therefore, the ET-GSNet can both extract long-distance dependency features and local structure features after model converge. Finally, as shown in Fig. 2, the student model with superior ability of

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extracting discriminative features is used to predict the final semantic class of samples in the phase of testing.

The proposed ET-GSNet method is summarized in Algorithm 1.

Algorithm 1 Proposed ET-GSNet Method

Training:

Stage 1: Training with small KD loss coefficient

Input: Training images, I_{in}; Temperature, T; KD loss coefficient, α; Teacher model, Pretrained ViT; Student model, Pretrained ResNet18; Function sets of teacher model, $f_{\text{tea}}()$; Function sets of student model, $f_{\text{stu}}()$;

Output: The optimized teacher model and student model;

$F_{\text{tea}} \leftarrow f_{\text{tea}}(I_{\text{in}})$;

$F_{\text{stu}} \leftarrow f_{\text{stu}}(I_{\text{in}})$;

$L_{\text{KD}} \leftarrow T^2 \cdot \text{KLdiv}\left(Q_{\text{stu}}^T, Q_{\text{tea}}^T\right)$; Q_{stu}^T and Q_{tea}^T are computed according to (18) and (19);

$L_{\text{fir}} \leftarrow \alpha L_{\text{KD}} + L_c^{\text{tea}} + L_c^{\text{stu}}$;

optimize teacher model and student model simultaneously with L_{fir};

Stage 2: Training with big KD loss coefficient

Input: Training images, I_{in}; Temperature, T; KD loss coefficient, α; Teacher model, ViT; Student model, ResNet18; Function sets of teacher model, $f_{\text{tea}}()$; Function sets of student model, $f_{\text{stu}}()$;

Output: The optimized student model;

$F_{\text{tea}} \leftarrow f_{\text{tea}}(I_{\text{in}})$;

$F_{\text{stu}} \leftarrow f_{\text{stu}}(I_{\text{in}})$;

$L_{\text{KD}} \leftarrow T^2 \cdot \text{KLdiv}\left(Q_{\text{stu}}^T, Q_{\text{tea}}^T\right)$; Q_{stu}^T and Q_{tea}^T are computed according to (18) and (19);

$L_{\text{sec}} \leftarrow 2\alpha L_{\text{KD}} + L_c^{\text{stu}}$;

optimize student model with L_{sec} and stop teacher model training;

Testing:

Scene classification

Input: Testing images, $I_{\text{in}}^{\text{test}}$; Student model, ResNet18; Function sets of student model, $f_{\text{stu}}()$;

Output: The predicted label Y_{label};

$F_{\text{stu}} \leftarrow f_{\text{stu}}\left(I_{\text{in}}^{\text{test}}\right)$;

$Y_{\text{label}} \leftarrow \text{softmax}(F_{\text{stu}})$;

The semantic label Y_{label} of each testing image is generated.

III. Experimental Results and Discussion

To evaluate the effectiveness of the proposed ET-GSNet method, five public remote sensing datasets are employed for exhaustive experiments, including parameter analysis, experimental comparison of different algorithms, confusion matrix analysis, visualization, and generalization analysis.

A. Experimental Setup

1) Datasets Description: In this manuscript, the Aerial Image Dataset (AID)[1] [6], the NWPU-RESISC45 Dataset (NWPU-RESISC45)[2] [7], the UC-Merced Land Use Dataset (UCM)[3] [79], and OPTIMAL-31[4] [51] are selected to conduct the following experiments. These four widely used datasets are challenging for scene classification, which possess the characteristics of high interclass similarity and intraclass diversity, and example images are shown in Figs. 5–8. Furthermore, more detailed information of the AID, NWPU-RESISC45, UCM, and OPTIMAL-31 datasets is shown in Table I.

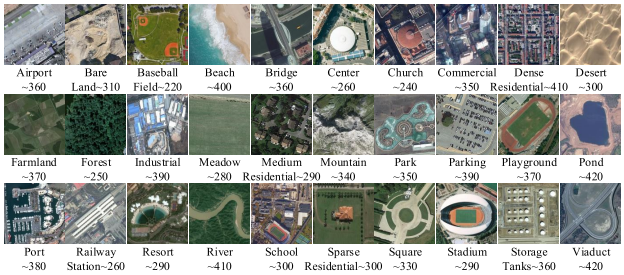


Fig. 5. One scene image per category of AID dataset is displayed (Class∼Number of images).

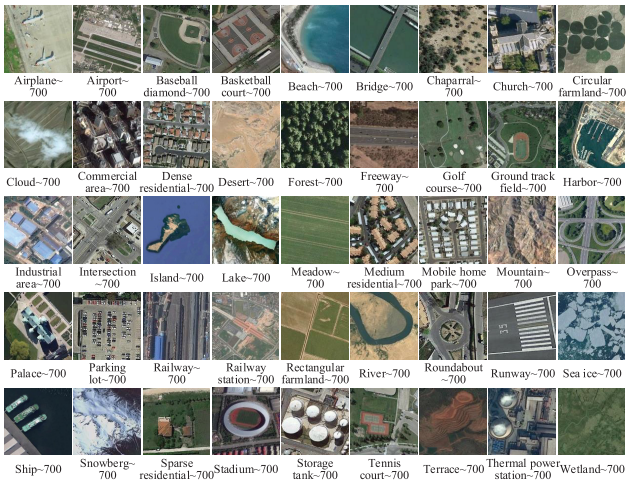


Fig. 6. One scene image per category of NWPU-RESISC45 dataset is displayed (Class∼Number of images).

TABLE I

Detail Information of AID, NWPU-RESISC45, OPTIMAL-31, and UCM Datasets

Data sets	AID	NWPU-RESISC45	OPTIMAL-31	UCM
Classes	30	45	31	21
Images per class	220∼420	700	60	100
Total images	10000	31500	1860	2100
Spatial resolution	0.5∼8	0.2∼30	–	0.3
Images size	600×600	256×256	256×256	256×256
Training ratios	20%&50%	10%&20%	80%	80%
Data source	Google Earth	Google Earth	Google Earth	USGS
Published year	2017	2017	2019	2010

2) Hardware and Software Environment: All experiments are carried out on a personal computer, and the detailed computing environment includes a 24-GB memory, an Intel Core i7-7800 CPU, and a NVIDIA TITAN Xp GPU. The software environment conducting the ET-GSNet is Python 3.7 with the Pytorch performing on the Windows 10 system.

3) Parameter Optimization Setup: The ViT model named "B_16_imagenet1k" from PyTorch-Pretrained-ViT package[5]

[1]http://www.lmars.whu.edu.cn/xia/AID-project.html

[2]http://www.escience.cn/people/JunweiHan/NWPU-RESISC45.html

[3]http://weegee.vision.ucmerced.edu/datasets/landuse.html

[4]http://crabwq.github.io/

[5]https://github.com/lukemelas/PyTorch-Pretrained-ViT

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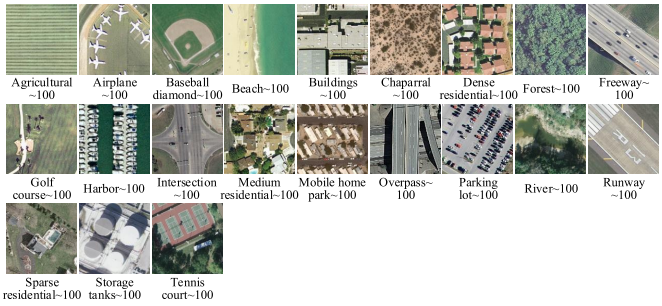


Fig. 7. One scene image per category of UCM dataset is displayed (Class∼Number of images).

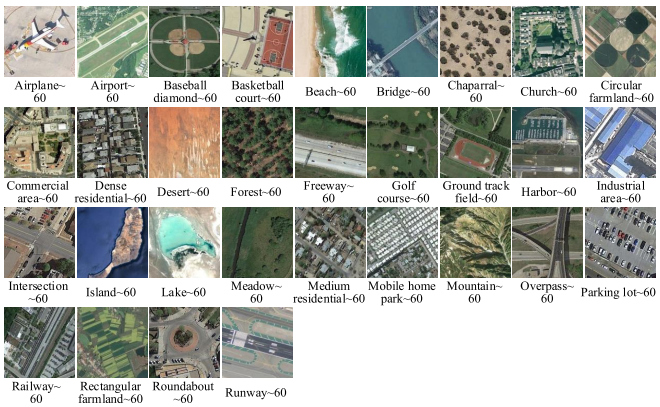


Fig. 8. One scene image per category of OPTIMAL-31 dataset is displayed (Class∼Number of images).

provides the pretrained weights of the teacher model (ViT), and initial parameters of the student model (ResNet18) are downloaded from the torchvision, which can be found on the official website of Pytorch. Due to the limitation of GPU, the batch size is set to 24, and all images are resized to 256×256 pixels before entering the ET-GSNet. The image patch size P_r of the teacher model (ViT) is set to 16×16, and the dimension of feature that produced from ViT is set to 512. For the teacher network, the stochastic gradient descent (SGD) is selected to optimize the ViT model until convergence, the corresponding momentum is set to 0.9, and the weight decay is 0.0001. As for the student network (ResNet18), the adopted optimizer is the Adam, and the parameter of weight decay is also set at 0.0001. The LR of each layer in the teacher model is set to 0.01, and the LR in the student model is set to 0.0001, except the last FC layer (0.001). In the stage of model training, the LR is divided by a factor of 10 for every 15 epochs.

4) Data Processing and Evaluation Metrics: Two kinds of data augmentation strategies, random flip and rotation, are adopted in the experiments. For the AID and NWPU-RESISC45 datasets, all images are randomly divided into the training set and testing set. The two training ratios for the AID dataset are 20% and 50%, and the proportion of training samples in the NWPU-RESISC45 is 10% and 20%. In the OPTIMAL-31 and UCM datasets, the training ratio is 80%. The abovementioned training ratios are the same with the works in [6], [7], [51], [79]. Considering the reliability of classification results, all experiments are repeated five times. Furthermore, the overall accuracy (OA) with standard deviation (STD) is employed to evaluate the

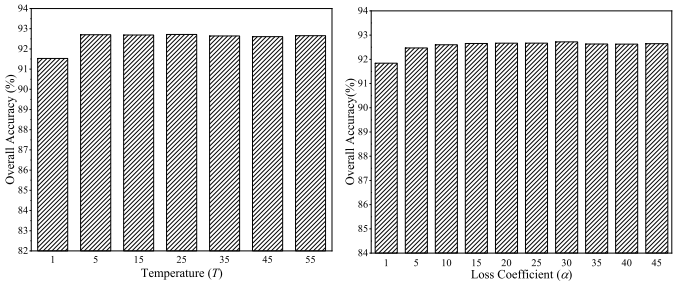


Fig. 9. OAs with respect to different values of (a) temperature and (b) KD loss coefficient on the NWPU-RESISC45 dataset under the training ratio of 10%.

classification performance of the proposed method on the four datasets. Another important evaluation metric is confusion matrix, which can reflect the detailed confusion degree and between-class classification errors.

B. Parameter Sensitivity Analysis

For the proposed ET-GSNet method, two free parameters related to the loss are crucial to determine the classification performance, including temperature (T) and KD loss coefficient (α). In this section, parameter sensitivity experiments are conducted on the NWPU-RESISC45 dataset (training ratio: 10%) to analyze their influence on the ET-GSNet method and select the optimal value. It is worth mentioning that the selected appropriate parameters of T and α are also applied to the other three datasets in the following experiments.

To validate the OAs with respect to different values of temperature, parameter T is tuned with a given set $\{1, 5, 15, \ldots, 55\}$, and variations of OAs concerning parameter T are shown in Fig. 9(a). As shown in Fig. 9(a), the OAs appear to increase dramatically at the beginning and then maintain stable when T varying between 5 and 55. This is because the student model is trained with hard labels under the circumstance of a small value of T (e.g., $T = 1$) and fails to distill long-range information of ViT. Furthermore, Fig. 9(a) shows that the ET-GSNet is able to obtain satisfactory classification results in a wide range of T, and the highest OA is achieved when T reaches 25. Therefore, parameter T is set to 25 in the following experiments on the NWPU-RESISC45, AID, OPTIMAL-31, and UCM datasets.

In the two-stage optimization of ET-GSNet, a KD loss coefficient is introduced to balance the importance of several different objective functions. To explore the performance of the proposed model with different weights, α is evaluated on the NWPU-RESISC45 dataset with a set of $\{1, 5, 10, \ldots, 40, 45\}$. According to Fig. 9(b), OAs first gradually improve with the increase of α and then reach a peak. When α exceeds a certain value, the trend of OAs is slightly declining. The reason is that an appropriate α can balance the importance of soft and hard labels well and promote the knowledge transferred from ViT to ResNet18. A small α cannot make the student model fully accept dark knowledge for weakening the weight of soft label, and the optimal α is set to 30 for four datasets.

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TABLE II
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE AID AND NWPU-RESISC45 DATASETS ($OA_{(STD)}(\%)$)

Method Type	Method	Backbone	Published Year	AID		NWPU-RESISC45	
				20%	50%	10%	20%
†	VGG-VD-16 [6], [7]	VGGNet	–	$86.59_{(0.29)}$	$89.64_{(0.36)}$	$76.47_{(0.18)}$	$79.79_{(0.15)}$
	VGG-VD16+MSCP [50]	VGGNet	2018	$91.52_{(0.21)}$	$94.42_{(0.17)}$	$85.33_{(0.17)}$	$88.93_{(0.14)}$
	DFAGCN [46]	VGGNet	2021	–	$94.88_{(0.22)}$	–	$89.29_{(0.28)}$
‡	MIDC-Net_CS [36]	DenseNet	2020	$88.51_{(0.41)}$	$92.95_{(0.17)}$	$86.12_{(0.29)}$	$87.99_{(0.18)}$
	RADC-Net [37]	DenseNet	2020	$88.12_{(0.43)}$	$92.35_{(0.19)}$	$85.72_{(0.25)}$	$87.63_{(0.28)}$
◊	Inception-v3-CapsNet [66]	Inception	2019	$93.79_{(0.13)}$	$96.32_{(0.12)}$	$89.03_{(0.21)}$	$92.60_{(0.11)}$
	IB-CNN(M) [63]	MobileNet	2020	$94.23_{(0.16)}$	$96.57_{(0.28)}$	$90.49_{(0.17)}$	$93.33_{(0.21)}$
	IB-CNN(R) [63]	ResNet	2020	$93.81_{(0.11)}$	$96.48_{(0.19)}$	$90.46_{(0.16)}$	$93.11_{(0.13)}$
	$GoogleNet_{global+local}$ [60]	GoogleNet	2020	$94.26_{(0.15)}$	$96.65_{(0.11)}$	$90.41_{(0.12)}$	$92.95_{(0.09)}$
	$ResNet18_{global+local}$ [60]	ResNet	2020	$94.38_{(0.10)}$	$96.76_{(0.20)}$	$90.04_{(0.15)}$	$92.79_{(0.11)}$
	MI-CNN multi-model fusion [61]	VGGNet	2020	$94.93_{(0.18)}$	$96.60_{(0.21)}$	$92.25_{(0.21)}$	$93.92_{(0.13)}$
	ResNet50+DOCP [64]	ResNet	2021	–	–	$89.74_{(0.21)}$	$91.84_{(0.33)}$
	MSA-Network [80]	ResNet	2021	$93.53_{(0.21)}$	$96.01_{(0.43)}$	$90.38_{(0.17)}$	$93.52_{(0.21)}$
	LSE-Net [59]	VGGNet	2021	$94.41_{(0.16)}$	$96.36_{(0.19)}$	$92.23_{(0.14)}$	$93.34_{(0.15)}$
	CSDS [81]	Xception	2021	$94.29_{(0.35)}$	$96.70_{(0.14)}$	$91.64_{(0.16)}$	$93.59_{(0.21)}$
	EFPN-DSE-TDFF [82]	ResNet	2021	$94.02_{(0.21)}$	$94.50_{(0.30)}$	–	–
	GLDBS [62]	ResNet	2021	$95.45_{(0.19)}$	$\mathbf{97.01}_{(0.22)}$	$92.24_{(0.21)}$	$94.46_{(0.15)}$
	RANet [83]	ResNet	2021	$92.71_{(0.14)}$	$95.31_{(0.37)}$	–	–
△	Proposed V16_21k [224×224] [72]	ViT	2021	$94.97_{(0.01)}$	–	$92.60_{(0.10)}$	–
★	**ET-GSNet**	**ResNet**	**Ours**	$\mathbf{95.58}_{(0.18)}$	$96.88_{(0.19)}$	$\mathbf{92.72}_{(0.28)}$	$\mathbf{94.50}_{(0.18)}$

†: Pretrained Model Based Methods ‡: Fully Training Based Methods ◊: Fine-tuning Based Methods △: ViT Based Methods ★: Proposed Methods

C. Comparison With Some State-of-the-Art Methods

To illustrate the advancement of the proposed method, some state-of-the-art (SOTA) methods are listed to compare with ET-GSNet on the two relatively large-scale datasets, including AID and NWPU-RESISC45. The comparison results are shown in Table II, where the training ratios of the AID dataset are 20% and 50%, and 10% and 20% training images of the NWPU-RESISC45 dataset are selected. For all listed methods, adopted training ratios are the same for a fair comparison. Furthermore, to clearly present the experiment results, all methods are separated into four categories, including pretrained model-based methods (†), fully training-based methods (‡), fine-tuning-based methods (◊), and ViT-based methods (△).

In Table II, VGG-VD-16 [6], [7] is regarded as the baseline of CNN-based methods. It is obvious that the pretrained model-based methods in [50] and [46] are superior to MI densely-connected convolution neural networks with channel-spatial attention (MIDC-Net_CS) [36] and residual attention based dense connected convolutional neural network (RADC-Net) [37]. The reason is that these fully training-based methods construct and train a model from scratch, and they are always limited by scarce labeled samples. Compared to the aforementioned methods, fine-tuning-based methods are more competitive because the pretrained model will be more appropriate for HSR images after fine-tuning. For listed fine-tuning-based methods, Inception-v3-CapsNet [66] combines CNN with CapsNet to preserve the spatial information of HSR images. Improved bilinear(IB)-CNN(M) and IB-CNN(R) [63] design IB pooling to effectively refine high-order and spatial orderless information. In $ResNet18_{global+local}$, $GoogleNet_{global+local}$ [60], global-local dual-branch structure (GLDBS) [62], and MI-CNN multimodel fusion [61], local information is fully considered and employed to boost the classification performance. To complement global scene-level features, ResNet50+DOCP [64] and enhanced feature pyramid network (EFPN)-deep semantic embedding (DSE)-two-branch deep feature fusion (TDFF) [82] extract object-level features via DOCP and EFPN for classification, respectively. For MSA-Network [80] and local semantic enhanced (LSE)-Net [59], they all concentrate on local key features perception to make the network focus on more salient regions. The attention mechanism is adequately employed in channel-spatial attention mechanism based on a depthwise separable convolution (CSDS) [81] and relation-attention networks (RANet) [83] to effectively exploit shallow and intermediate features of CNNs. Compared to the aforementioned CNN-based methods, the listed ViT-based method (i.e., Proposed V16_21k [224 × 224] [72]) can achieve higher classification results. Although the method successfully explores long-distance dependencies among image patches, important local information is ignored, which limits the improvement of classification performance. In addition, due to big model size, long computation time for the ViT-based method is an inevitable issue.

According to Table II, our proposed method obtains competitive results on the AID and NWPU-RESISC45 datasets compared to most SOTA methods. Specifically, the highest OAs of AID (training ratio: 20%) and NWPU-RESISC45 (training ratio: 10% and 20%) are achieved by the ET-GSNet method. It is worth mentioning that the ET-GSNet merely employs small parameters network (i.e., ResNet18) and even better than other methods with complex models. For the listed SOTA methods in Table II, the GLDBS [62] obtains the highest OA on the AID under the training ratio of 50%, and this is because GLDBS adopts dual-branch network as backbones, which combines ResNet18 and ResNet34. Although the proposed ET-GSNet method does not reach the highest accuracy on the AID dataset under 50% training samples, it also ranks second among all methods. As a result, the ET-GSNet has

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TABLE III

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE OPTIMAL-31 AND UCM DATASETS ($OA_{(STD)}(\%)$). THE TRAINING RATIO OF TWO DATASETS IS BOTH 80%

Method	Backbone	Year	OPTIMAL-31	UCM
ARCNet-VGGNet16 [51]	VGGNet	2019	$92.70_{(0.35)}$	$99.12_{(0.14)}$
ResNet_LGFFE [84]	ResNet	2019	$94.55_{(0.36)}$	$98.62_{(0.88)}$
EfficientNet-B3-aux [85]	EfficientNet	2019	$94.51_{(0.75)}$	$99.09_{(0.17)}$
DenseNet+FPN [86]	DenseNet	2020	$95.23_{(0.25)}$	–
GBNet+global feature [87]	VGGNet	2020	$93.28_{(0.27)}$	$98.57_{(0.48)}$
MAA-CNN [88]	ResNet	2020	$95.70_{(0.54)}$	–
KFBNet(DenseNet-121) [19]	DenseNet	2020	$95.60_{(0.63)}$	$\mathbf{99.88}_{(0.12)}$
$ResNet18_{global+local}$ [60]	ResNet	2020	–	$99.52_{(0.24)}$
GRMA-Net-ResNet18 [67]	ResNet	2021	$96.23_{(1.56)}$	$99.52_{(0.13)}$
ResNet50+DOCP [64]	ResNet	2021	–	$99.15_{(0.40)}$
Proposed V16_21k[224×224] [72]	ViT	2021	$95.07_{(0.12)}$	–
RANet [83]	ResNet	2021	$94.61_{(0.18)}$	$99.27_{(0.24)}$
ET-GSNet	**ResNet**	**Ours**	$\mathbf{96.45}_{(0.77)}$	$99.29_{(0.34)}$

excellent ability of feature extraction, and it can effectively learn more potential information from the teacher model without sacrificing computation time in the testing stage.

To comprehensively evaluate the superiority of the proposed ET-GSNet method, two relatively small-scale datasets, OPTIMAL-31 and UCM datasets, are also selected to perform a comparison with some SOTA methods, and Table III demonstrates the comparison results. All listed methods on two datasets are implemented under the training ratio of 80%, which is the same as previous works [51], [79].

In Table III, most comparison algorithms are considered ResNet as backbone, which is consistent with the proposed method. In addition, SOTA methods based on other networks are also considered to compare the classification performance, including ViT, DenseNet, EfficientNet, and VGGNet. Among these methods, attention recurrent convolutional network (ARCNet)-VGGNet16 [51], multiaugmented, attention-based CNN (MAA-CNN) [88], gated recurrent multiattention neural network (GRMA-Net)-ResNet18 [67], and key filter bank (KFB)Net [19] are all based on attention mechanism to highlight informative regions in the HSR images. EfficientNet-B3-aux [85] and gated bidirectional network (GBNet)+global feature [87] employ auxiliary loss function and hierarchical feature aggregation, respectively. $ResNet18_{global+local}$ [60] and ResNet_local-global-fusion feature extraction (LGFFE) [84] focus on exploring local and global features to obtain the discriminative feature representation. DenseNet+feature pyramid networks (FPN) [86] and ResNet50+DOCP [64] boost the classification performance through the aid of key objects location.

From Table III, compared to some SOTA methods, the ET-GSNet still maintains satisfactory results on the two relatively small-scale datasets. For the OPTIMAL-31 dataset, the highest classification accuracy is obtained by our proposed method, and OA reaches 96.45%. According to Table III, on the UCM dataset, OA from the proposed method is slightly less than those from $ResNet18_{global+local}$ and GRMA-Net-ResNet18, but the ET-GSNet achieves better results on the other three datasets (i.e., AID, NWPU-RESISC45, and OPTIMAL-31). Although KFBNet(DenseNet-121) achieves the best classification performance on the UCM dataset, the classification result of

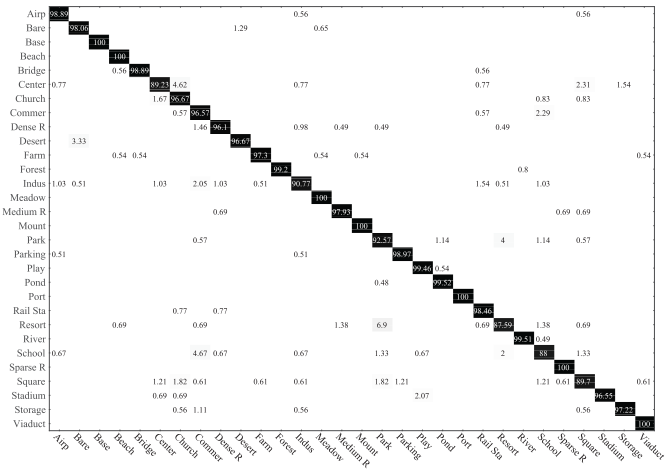


Fig. 10. Confusion matrix obtained through the ET-GSNet method on the AID dataset. (Training ratio: 50%.)

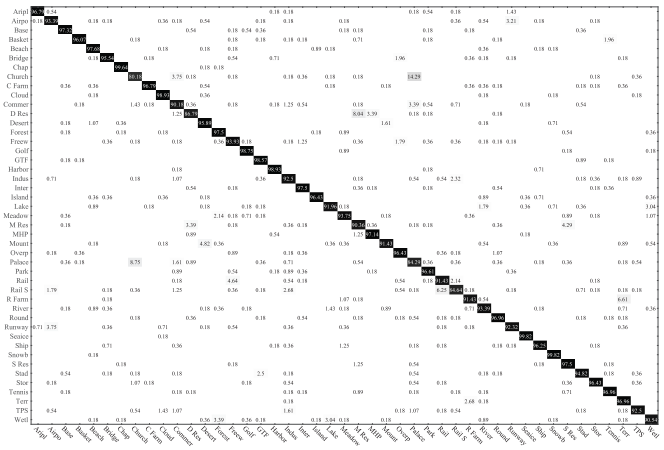


Fig. 11. Confusion matrix obtained through the ET-GSNet method on the NWPU-RESISC45 dataset. (Training ratio: 20%.)

the ET-GSNet is still praiseworthy. This may be attributed to the accuracy of small-scale dataset that is saturated, and the difference in accuracy among different methods is quite small (within 0.6%). Furthermore, the teacher model of the ET-GSNet possesses large parameters, while the UCM dataset has small scale of image numbers, and these factors limit the accuracy improvement of the proposed method.

D. Confusion Matrix Analysis

To further intuitively evaluate confusion degree and between-class classification errors, we provide two confusion matrices on two datasets by using the proposed ET-GSNet method. Figs. 10 and 11 show the experimental results on the AID (50% training samples) and NWPU-RESISC45 (20% training samples) datasets, respectively.

As shown in Fig. 10, impressive performance is reflected on the AID dataset by employing the ET-GSNet strategy. Specifically, accuracies of 26 categories are more than 90%, and the remaining classes are all over 87%. Furthermore, total seven semantic classes reach 100%, including "baseball field," "beach," "meadow," "mountain," "port," "sparse residential," and "viaduct". This is because the developed method effectively combines local features and long-distance features for precise classification. Due to the similar ground objects, the

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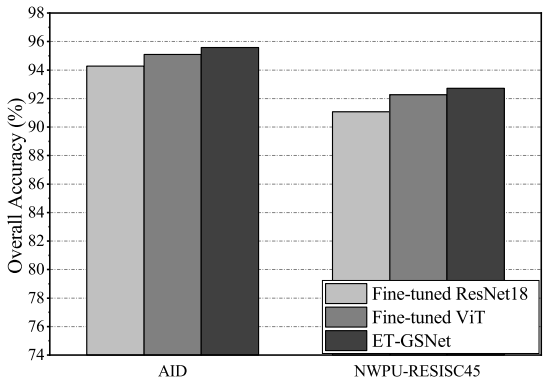


Fig. 12. Ablation study of different methods on two datasets. (AID: 20% training samples and NWPU-RESISC45: 10% training samples.)

classes of "center," "resort," "school," and "square" are easy to be confused. For these artificial categories, the ET-GSNet model can explore a spatial structure via learning knowledge of long-range dependency from ViT, and it reduces confusion to a great extent.

Fig. 11 shows the confusion matrix obtained by the proposed ET-GSNet on the NWPU-RESISC45 dataset. As a challenging dataset, satisfactory classification results also generate with an ET-GSNet method (all over 80%). Some categories, such as "chaparral," "sea ice," and "snowberg," almost achieve 100% classification accuracy. The reason is that the proposed ET-GSNet employs KD to excavate the potential dependency in HSR images. For the NWPU-RESISC45 dataset, the main confusion occurs on "church," "palace," and "railway station" classes for the same components of scene (e.g., trees, buildings, and roads). The ET-GSNet captures competitive classification results on such classes, which is benefitted from the outstanding learning ability of student model, and it further improves the discriminative capacity of features.

E. Ablation Study and Analysis of Computation Consumption

To evaluate the effectiveness of different components, the ablation study of teacher model (i.e., fine-tuned ViT) and student model without and after KD (i.e., fine-tuned ResNet18 and ET-GSNet) is organized on the AID and NWPU-RESISC45 datasets, and the OAs obtained by different methods are shown in Fig. 12. In the ablation experiments, the training samples of the AID account for 20%, while the training ratio of the NWPU-RESISC45 dataset is 10%.

From Fig. 12, it is obvious that the fine-tuned ViT can achieve better performance than fine-tuned ResNet18 on both datasets, and the reason is that ViT possesses the outstanding ability of extraction context features in HSR images via self-attention. Compared to fine-tuned ResNet18, the ET-GSNet improves 1.31% and 1.66% on the AID and NWPU-RESISC45 datasets in terms of OAs, respectively. This is because the ET-GSNet comprehensively accepts and absorbs long-range dependency information from ViT through KD, which is critical to further enhancement of classification accuracy. In addition, the classification performance of the ET-GSNet is even surpassing that of the teacher model on two datasets. It may account for the proposed ET-GSNet that further boosts the discriminative ability of features via merging local information and long-range relationship.

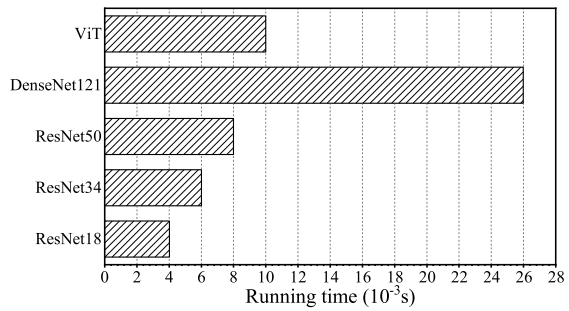


Fig. 13. Running time comparison on the NWPU-RESISC45 dataset. The running results reported in this histogram are forward propagation time of samples on one batch, and the employed computer is the same.

For the ET-GSNet, the gap of parameters between the student model and the teacher model is large, which is closely related to computation time. To quantitatively analyze the operation efficiency of the proposed method, the running time of different methods conducted on the NWPU-RESISC45 dataset is shown in Fig. 13, and they are calculated on the same computer for a fair comparison. It has to state that the meaning of running time is forward propagation testing time for a minibatch, which includes 24 samples.

As shown in Fig. 13, DenseNet121 consumes the longest time among all methods because it possesses a deep network structure with million parameters. As the teacher model of ET-GSNet method, the running time of ViT is larger than that of all ResNets. This is because the model size of ViT is bigger than all ResNets for complex matrix calculation. Furthermore, for the rest of the methods (i.e., ResNet18, ResNet34, and ResNet50), computation time gradually increases with the deepening of network depth. Obviously, the shortest time is consumed by ResNet18, which is chosen as the student model of ET-GSNet in the testing stage. Therefore, the proposed method is more efficient in the model inference and can save computing resources in terms of time and hardware.

F. Visualization Experiment

For HSR images, scene labels are highly related to key areas and objects, and accurately locating the attention regions is conducive to improve classification results. To reflect the accuracy of location based on the ET-GSNet, the energy map of some samples selected from different categories of the NWPU-RESISC45 is listed in Fig. 14, and the fine-tuned ResNet18 is considered as the comparison method. In Fig. 14, the red region highlights the key area relative to semantic, and the blue region corresponds to the irrelative areas that have to be suppressed.

As shown in Fig. 14, it is clear that the distribution of attention regions produced via the ET-GSNet is more accurate than the fine-tuned ResNet18. The area located by the ET-GSNet is concentrated and well-defined and thus can precisely predict semantic labels. Taking "basketball court" as an example, the ET-GSNet can successfully focus on the basketball court even if it locates in the corner of the HSR image, while the fine-tuned ResNet18 concerns some

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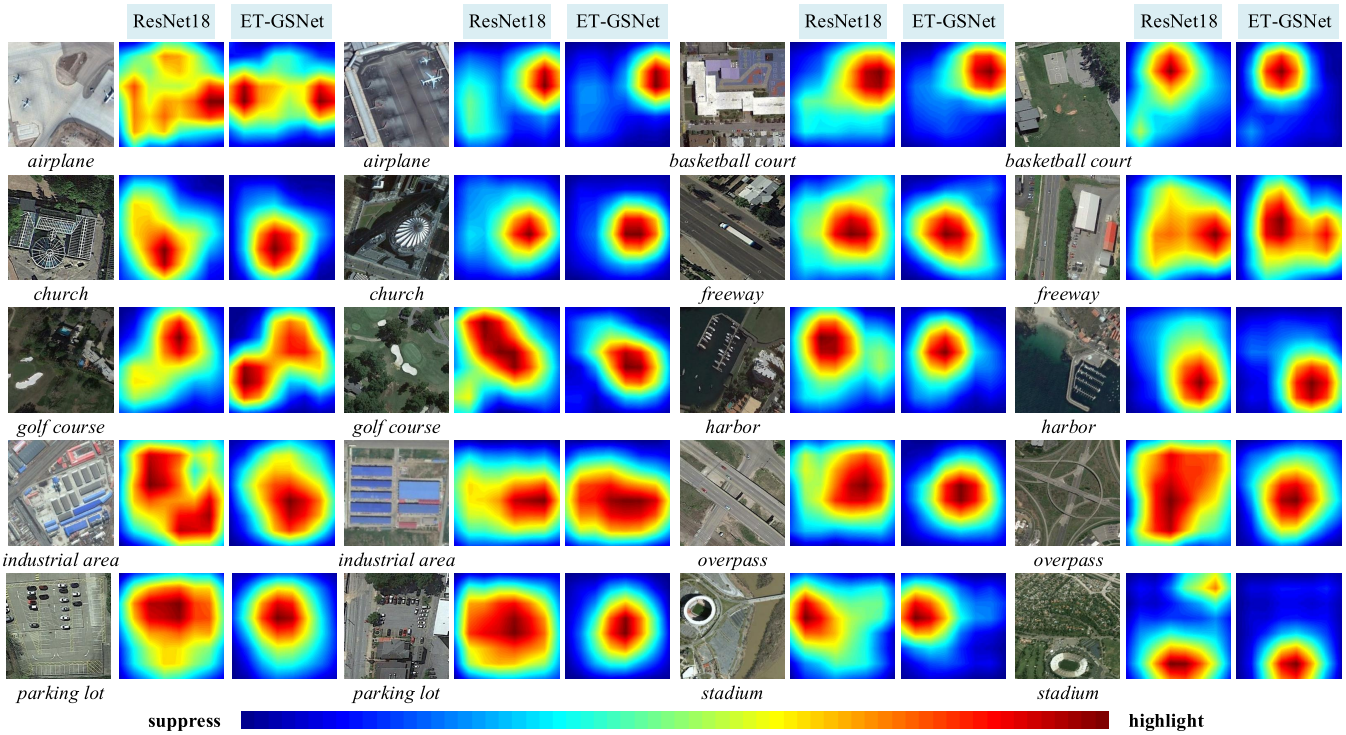


Fig. 14. Some samples of energy maps based on the proposed ET-GSNet and the fine-tuned ResNet18 in the NWPU-RESISC45 dataset.

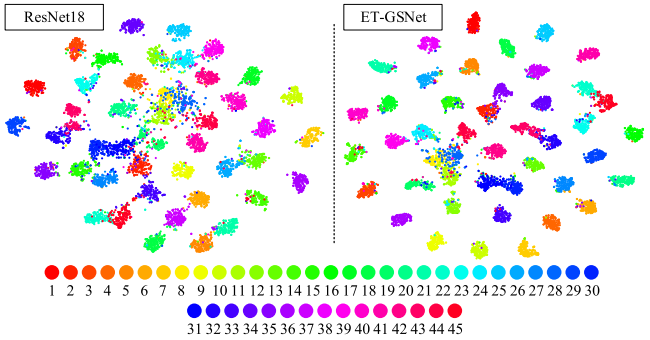


Fig. 15. Two-dimensional embedding of ResNet18 and the ET-GSNet in the NWPU-RESISC45 dataset under the training ratio of 10%.

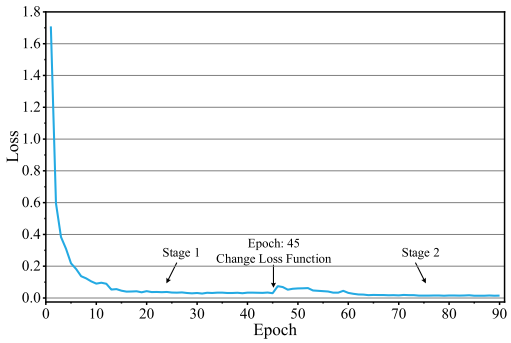


Fig. 16. Training loss of student model in the ET-GSNet method, and 10% samples of the NWPU-RESISC45 dataset are used for model optimization.

insignificant areas, such as buildings. In addition, the bound of fine-tuned ResNet18 is fuzzy, which is not conducive to improve the classification performance. For indistinguishable class "church," the ET-GSNet can concentrate on the crucial object and effectively reduce the confusion, which is consistent with results in Fig. 11. Furthermore, the same conclusion can also be summarized in the remaining categories, and it is because the proposed ET-GSNet based on KD learns long-range dependency information and local structure feature from HSR images at the same time.

To intuitively compare the separability of features extracted from ResNet18 and the ET-GSNet, the t-stochastic neighbor embedding (SNE) method [89] is employed to visualize two-dimensional embedding on the NWPU-RESISC45 dataset. In the experiment, 10% training samples are adopted to optimize models, and 200 samples per class in the testing set are randomly selected to visualize the two-dimensional embedding. Fig. 15 shows the data distributions of two models.

From Fig. 15, sample distribution obtained by ResNet18 tends to overlap between different classes, and this phenomenon may have resulted from the limited discrimination ability of the method. Compared to ResNet18, it is obvious that the proposed ET-GSNet method greatly alleviates the confusion and achieves good classification results on the NWPU-RESISC45 dataset. Specifically, the proposed method increases the separability and relative distance of data points from different classes, and the distribution of points in clusters from the same category becomes more compact. This is because the ET-GSNet can effectively extract scene-level features and represent the abstract semantic information of HSR images.

To evaluate the convergence of the proposed ET-GSNet method, the training loss of student model is visualized by using 10% scenes of the NWPU-RESISC45 dataset as training samples. The details are shown in Fig. 16.

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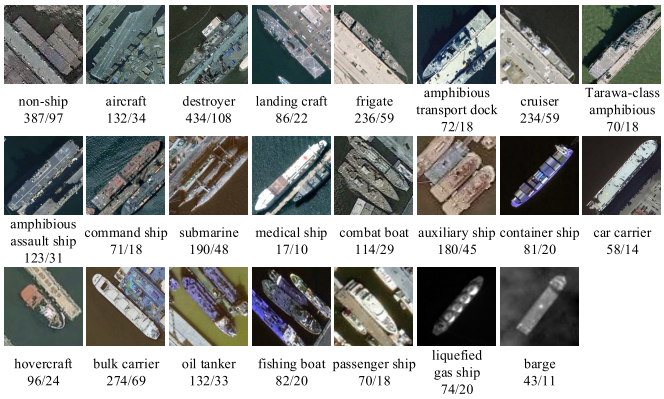


Fig. 17. One scene image per category of FGSC-23 dataset is displayed (number of train images/number of test images).

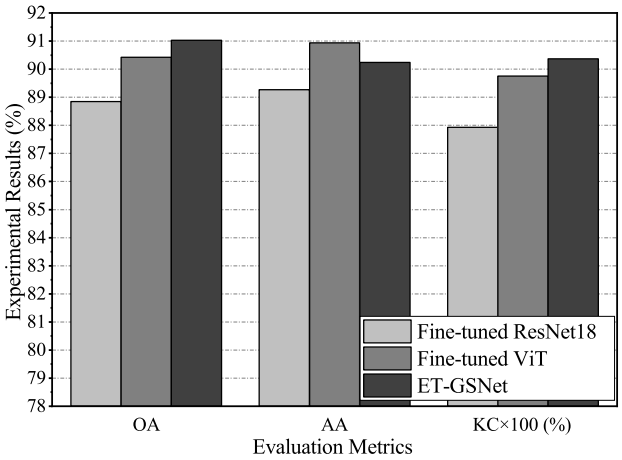


Fig. 18. Ablation results of OA, AA, and KC × 100 (%) on the FGSC-23 dataset.

As shown in Fig. 16, a loss value can fall relatively quickly, and the shock is not obvious. After the 45th epoch, the loss function changes from (14) to (15), and the loss of student model can be further reduced until convergence.

G. Generalization Analysis of the Proposed Method

To further validate the generalization ability of the ET-GSNet, the Fine-Grained Ship Collection-23 (FGSC-23) dataset that oriented in the fine-grained object recognition is adopted to demonstrate the adaptability of the proposed method to different types of scenarios.

1) Datasets Description: The FGSC-23 dataset[6] [90] is composed of over 4000 high-resolution optical remote sensing images, which are collected from Google Earth and Gaofen (GF)-2 satellite. As shown in Fig. 17, all images are divided into 23 categories, and the image resolution ranges from 0.4 to 2 m. In this article, the separation of training set and testing set is consistent with [90]. The direction of ship in the image is arbitrary while keeping the target center in line with the image center. The challenges of FGSC-23 dataset are category diversity, image scene diversity, instance label diversity, and category imbalance.

2) Parameters and Evaluation Metrics: All parameters setup are the same as in Section III-A, and it is worth mentioning that all experiments are only operated once. According to the results of parameter analysis, the parameters of T and α are set to 25 and 30, respectively. To comprehensively illustrate the classification results, OA, average accuracy (AA), kappa coefficient (KC), and confusion matrix are selected as the evaluation metrics in this section for quantitative analysis.

3) Ablation Results Analysis: The ablation results of the fine-tuned ResNet18, the fine-tuned ViT, and the ET-GSNet are shown in Fig. 18 based on the metrics of OA, AA, and KC × 100 (%). As shown in Fig. 18, in terms of OA and KC × 100 (%), the ET-GSNet is superior to the fine-tuned ResNet18 and fine-tuned ViT. The OA and KC × 100 (%) of the ET-GSNet improve 2.18% and 2.44% contrast to the fine-tuned ResNet18, respectively. This is benefitted from the extra knowledge that distilled from the ViT, and more discriminative features can be extracted to describe the fine-grained object scene images. In Fig. 18, it can be found that although the fine-tuned ViT obtained the highest result of AA among all methods, the AA of the ET-GSNet is still higher than that of the fine-tuned ResNet18. The reason is that the FGSC-23 dataset is class-imbalanced and the AA criteria are more sensitive to the number of every class, leading to individual samples that can fluctuate widely on the results of AA. Furthermore, the ET-GSNet shows more satisfactory performance on AA than the fine-tuned ResNet18, which indicates that the proposed method shows robust performance when applied to scenario data of different natures and masters outstanding generalization ability.

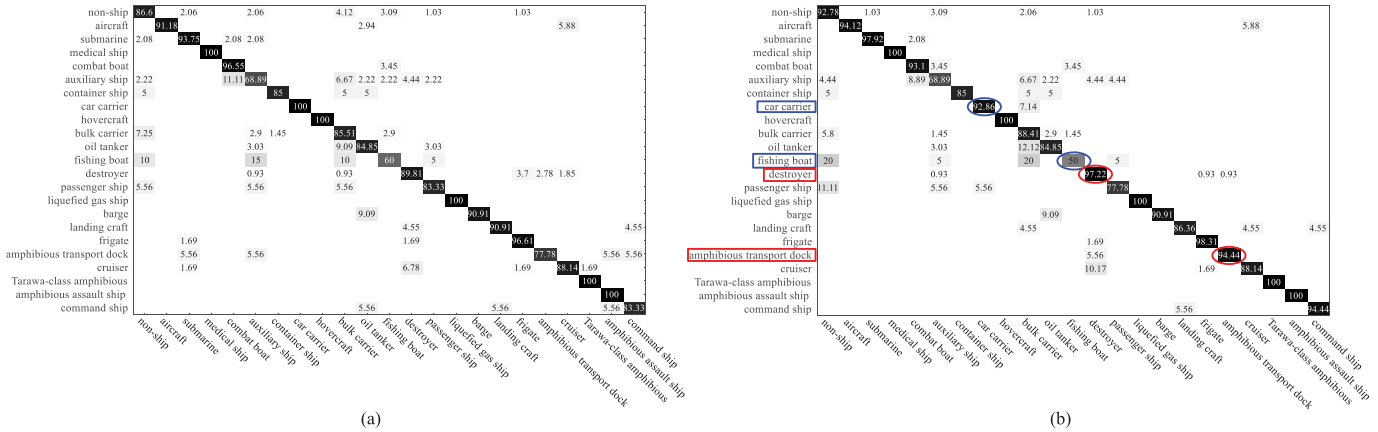
4) Confusion Matrix Analysis: The fine-grained recognition results of the fine-tuned ResNet18 and the ET-GSNet are shown in Fig. 19. In view of the difference between the two confusion matrices, it is obvious that the classification results of most classes (18 classes) are improved via the proposed method, such as "destroyer" and "amphibious transport dock". Specifically, compared to the fine-tuned ResNet18, the improvement amplitude of "destroyer" and "amphibious transport dock" is 7.41% and 16.66% in terms of OA, respectively. This is because the ResNet18 assimilates the long-distance dependence information well from the ViT model via KD, and it is conducive to improve the classification performance. From Fig. 19, although the experimental results of remaining classes (e.g., "car carrier" and "fishing boat") of the ET-GSNet are inferior to those of the fine-tuned ResNet18, the range of accuracy reduction is acceptable on the FGSC-23 dataset. When digging into the characteristics of the dataset, the phenomenon accounts for that the number of test samples in each class is imbalanced. Taking the class of "fishing boat" as an example, it possesses 20 testing samples, and a label prediction error results in a significant decrease (5%) in OA.

IV. Conclusion

In this article, a novel end-to-end method termed ET-GSNet is proposed for remote sensing scene classification. In the ET-GSNet method, ViT is adopted as the teacher model to extract abundant long-range contextual information, and it is transmitted to the student network (ResNet18) via KD.

[6]https://pan.baidu.com/s/1h_F7c-btLqhOxLT20XHWBg

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(a) (b)

Fig. 19. Confusion matrices on the FGSC-23 dataset. (a) Results obtained via the fine-tuned ResNet18. (b) Results obtained via the ET-GSNet. (The blue and red markers, respectively, highlight the two descending categories and two increasing categories, which are compared to the fine-tuned ResNet18.)

In the training stage, two networks are optimized together, and the LR of teacher model gradually decreases until zero, while the weight coefficient of KD loss is doubled. As a result, the student model in the ET-GSNet method possesses the abilities to capture long-distance dependencies and local geometric structure in remote sensing images. Experiments on the AID, NWPU-RESISC45, OPTIMAL-31, and UCM datasets indicate that the proposed ET-GSNet method has more competitive classification accuracies compared to some SOTA methods. Furthermore, results on the FGSC-23 dataset show that the proposed method possesses good generalization on different classification tasks. For the future work, we will try to reduce the time consumption of the ET-GSNet method in the training stage and further improve the performance on small-scale datasets.

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